8. Consider a Library Management System. Develop a JavaScript program that will validate the controls in the forms you have created for the application. State the assumptions you make (business logic you are taking into consideration).

Aim:

To develop a web-based **Library Management System** that validates user inputs using JavaScript and connects to a database using **Servlets and JSP**, while also demonstrating a simple **Weather Information Webpage UI**.

Procedure:

1. Create an HTML webpage for Library Management System.
2. Design a form to add books/users.
3. Apply JavaScript validation for all fields.
4. Create a JSP page to display data.
5. Develop a Servlet to handle form submission.
6. Connect the application to a MySQL database.
7. Insert validated data into the database.
8. Run the project on a server (e.g., Apache Tomcat).
9. Display success or error messages.

Code:

1.Html

```
<!DOCTYPE html>
<html>
<head>
  <title>Library Management System</title>
  <script>
    function validateForm() {
      let name = document.forms["libForm"]["name"].value;
```

```
    let email = document.forms["libForm"]["email"].value;
    let phone = document.forms["libForm"]["phone"].value;
    let book = document.forms["libForm"]["book"].value;

    let namePattern = /^[A-Za-z ]+$/;
    let emailPattern = /^[^ ]+@[^ ]+\.[a-z]{2,3}$/;

    if (name == "" || email == "" || phone == "" || book == "") {
        alert("All fields must be filled out");
        return false;
    }

    if (!name.match(namePattern)) {
        alert("Name must contain only alphabets");
        return false;
    }

    if (!email.match(emailPattern)) {
        alert("Invalid email format");
        return false;
    }

    if (phone.length != 10 || isNaN(phone)) {
        alert("Phone must be 10 digits");
        return false;
    }

    return true;
}
```

```
    </script>
</head>
<body>
    <h2>Library Management System</h2>
        <form name="libForm" action="LibraryServlet" method="post" onsubmit="return validateForm()">
        Name: <input type="text" name="name"><br><br>
        Email: <input type="text" name="email"><br><br>
        Phone: <input type="text" name="phone"><br><br>
        Book Name: <input type="text" name="book"><br><br>
        <input type="submit" value="Submit">
    </form>
</body>
</html>
```

2.Servlet (Java):

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;
import java.sql.*;

public class LibraryServlet extends HttpServlet {
    public void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

        String name = request.getParameter("name");
        String email = request.getParameter("email");
        String phone = request.getParameter("phone");
        String book = request.getParameter("book");
```

```
        try {
            Class.forName("com.mysql.jdbc.Driver");
            Connection con = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/library", "root", "password");

            PreparedStatement ps = con.prepareStatement(
                "INSERT INTO users(name,email,phone,book) VALUES(?,?,?,?)");

            ps.setString(1, name);
            ps.setString(2, email);
            ps.setString(3, phone);
            ps.setString(4, book);

            int i = ps.executeUpdate();

            response.setContentType("text/html");
            PrintWriter out = response.getWriter();

            if (i > 0) {
                out.println("<h3>Record Inserted Successfully!</h3>");
            } else {
                out.println("<h3>Error inserting record</h3>");
            }
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}
```

3. JSP Page (Display):

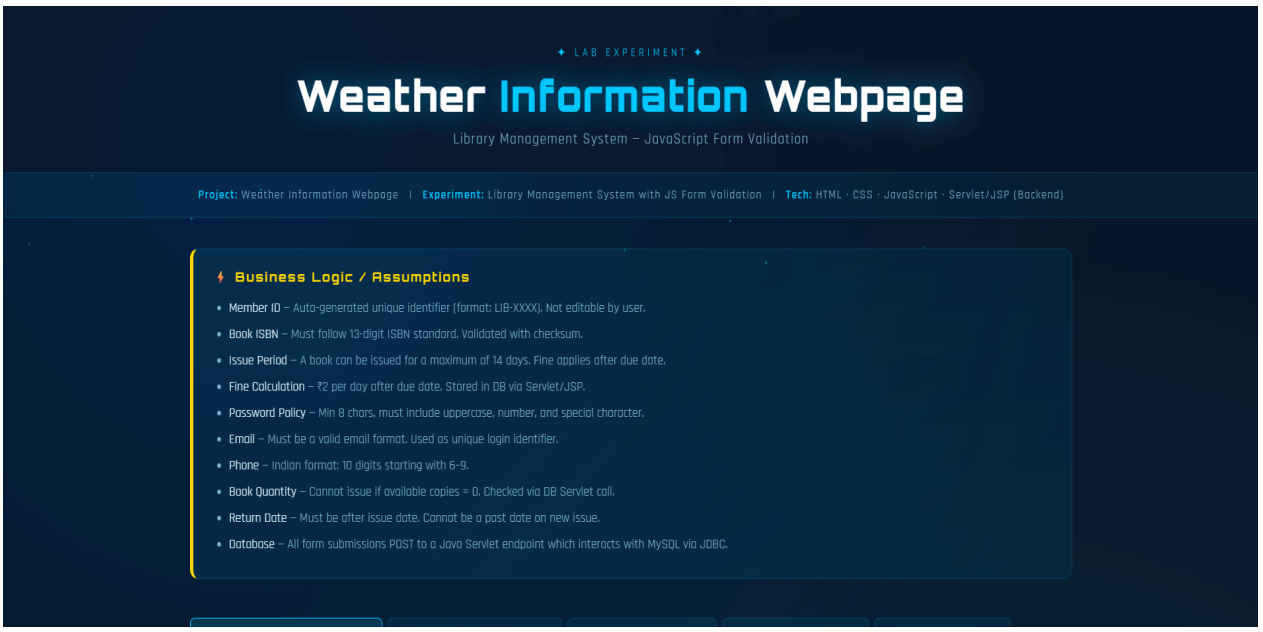
```
<%@ page import="java.sql.*" %>
<html>
<body>
<h2>Library Records</h2>
<table border="1">
<tr>
<th>Name</th>
<th>Email</th>
<th>Phone</th>
<th>Book</th>
</tr>
<%
try {
   Class.forName("com.mysql.jdbc.Driver");
   Connection con = DriverManager.getConnection(
      "jdbc:mysql://localhost:3306/library", "root", "password");
   Statement stmt = con.createStatement();
   ResultSet rs = stmt.executeQuery("SELECT * FROM users");
   while(rs.next()) {
%>
<tr>
<td><%= rs.getString("name") %></td>
<td><%= rs.getString("email") %></td>
<td><%= rs.getString("phone") %></td>
<td><%= rs.getString("book") %></td>
</tr>
<%
   }
```

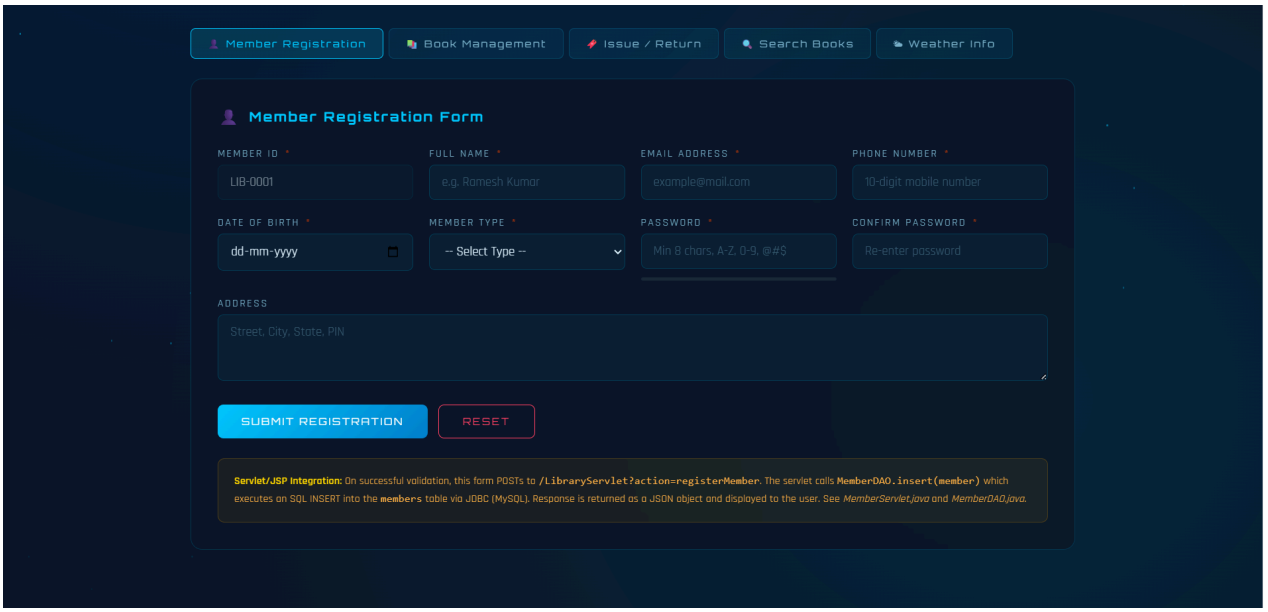
```
} catch(Exception e) {
    out.println(e);
}
%>
</table>
</body>
</html>
```

4. Simple Weather Webpage (UI only):

```
<!DOCTYPE html>
<html>
<head>
    <title>Weather Info</title>
</head>
<body>
    <h1>Weather Information</h1>
    <p>City: Chennai</p>
    <p>Temperature: 32°C</p>
    <p>Condition: Sunny</p>
</body>
</html>
```

Output:


✦ LAB EXPERIMENT ✦
Weather Information Webpage
Library Management System – JavaScript Form Validation
Project: Weather Information Webpage | Experiment: Library Management System with JS Form Validation | Tech: HTML · CSS · JavaScript · Servlet/JSP (Backend)
Business Logic / Assumptions
Member ID – Auto-generated unique identifier (format: LIB-XXXX). Not editable by user.
Book ISBN – Must follow 13-digit ISBN standard. Validated with checksum.
Issue Period – A book can be issued for a maximum of 14 days. Fine applies after due date.
Fine Calculation – ₹2 per day after due date. Stored in DB via Servlet/JSP.
Password Policy – Min 8 chars, must include uppercase, number, and special character.
Email – Must be a valid email format. Used as unique login identifier.
Phone – Indian format: 10 digits starting with 6–9.
Book Quantity – Cannot issue if available copies = 0. Checked via DB Servlet call.
Return Date – Must be after issue date. Cannot be a past date on new issue.
Database – All form submissions POST to a Java Servlet endpoint which interacts with MySQL via JDBC.


Member Registration
Book Management
Issue / Return
Search Books
Weather Info
Member Registration Form
MEMBER ID *
LIB-0001
FULL NAME *
e.g. Ramesh Kumar
EMAIL ADDRESS *
example@mail.com
PHONE NUMBER *
10-digit mobile number
DATE OF BIRTH *
dd-mm-yyyy
MEMBER TYPE *
-- Select Type --
PASSWORD *
Min 8 chars, A-Z, 0-9, @#$
CONFIRM PASSWORD *
Re-enter password
ADDRESS
Street, City, State, PIN
SUBMIT REGISTRATION
RESET
Servlet/JSP Integration: On successful validation, this form POSTs to /LibraryServlet?action=registerMember. The servlet calls MemberDAO.insert(member) which executes an SQL INSERT into the members table via JDBC (MySQL). Response is returned as a JSON object and displayed to the user. See MemberServlet.java and MemberDAO.java.

Result:

Thus, the **Library Management System** with JavaScript validation and database connectivity using **Servlet and JSP** was successfully developed and executed.